Zhuohan FANG

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EDUCATION

UC San Diego, School of Global Policy and Strategy 09/2022-06/2024

Master of Chinese Economic and Political Affairs (Environmental Policy Focus); **GPA: 3.67 / 4.0**

Core Modules: Quantitative Methods, Causal Inference, International Economics, GIS & Spatial Data Analysis, Remote Sensing, Environment & Regulation Economics, Energy Systems & Innovation, Chinese Politics

Honors: Full Scholarship; Dean's Fellow; Spatial Analysis Certificate; Quantitative Method Certificate

Zhejiang University 09/2018-06/2022

BSc Agricultural Economy; **GPA: 3.74 / 4.0**

Core Modules: Probability and Statistics, Linear Algebra, Calculus, Econometrics, Macroeconomics, Intermediate Microeconomics, Development Economics, Agricultural Economics, Public Economics, Finance

Honors: Merit Dissertation; Horner Program by Chu Kochen College & School of Public Administration

RESEARCH INTERESTS

Global climate governance as public goods provision;

Political economy of energy and environment; industrial and technology policy;

Clean energy transition and electricity market reform; US-China cooperation on sustainable development

WORK EXPERIENCE

Analyst | The Lantau Group (TLG), Shanghai Office 07/2025-Present

- Managed and maintained multiple energy and electricity and energy databases relevant to tariff structure, trading results, fuel prices and load profiles, ensuring regular updates to support China electricity market research
- Collected and analyzed China mainland and Taiwan market and policy input for calibrating energy cost and dispatchment in-house forecast models
- Engaged in ESG consulting for various clients, supporting projects like green Power Purchase Agreement (PPA) tendering, market environment research for clean energy manufacturer, and usage forecast for end-users

Staff Research Associate | Scripps Institution of Oceanography - SoCal Heat Hub 07/2024-06/2025

PI: Prof. Morgan Levy - Quantifying the Water Demand of Urban Green Spaces for Heat Adaptation in Arid Climates: Insights from Southern California

- Explored greening as a heat adaptation strategy, with emphasis on the ecohydrology of urban greening, water availability, and the sustainability of green solutions to climate change
- Extracted and analyzed pixel-level data from satellite imageries to evaluate the relationship between vegetation, water, and temperature, modelling mitigation effects from vegetation to extreme heat
- Designed and coded a public interactive web application that allows users to generate custom maps and perform graphical analyses of satellite geospatial data, and presented the interactive map to K-12 educators.

Staff Research Associate | UC Institute on Global Conflict and Cooperation (IGCC) 07/2024-06/2025

PI: Prof. Barry Naughton - China's Innovation Self-Reliance Goal: How Much Progress?

- Conducted analysis of China's industrial development and science & technology policy (STIP), focusing on technological self-reliance and its impact on import dependency, especially for bottleneck technologies
- Compiled and compared lists of "chokepoint" technologies from multiple resources to identify high-tech components by industry sectors
- Pulled out trade flow data from the Chinese National Customs to evaluate the outcome of China's self-reliance-oriented industrial policy, partitioned key patterns like stockpiling, domestic substitution, and import dependency

Graduate Student Researcher | Power Transformation Lab 07/2023-06/2024

PI: Prof. Teevrat Garg & Prof. Michael Davidson - Agenda on Clean Energy Trade between India and China

- Conducted data-driven research on trade patterns between India and China in clean energy commodity trade and the implications for global decarbonization efforts

- Constructed categories and datasets for clean energy commodity trade on HS8 level to assess the effectiveness of India's trade policies. Identified policy impacts of tariffs and domestic initiatives on manufacturing capacity and market dynamics under geopolitical responses like "de-risking" and "friend-shoring"
- Explored China's role in clean energy manufacturing GVC, examined exports and imports for different critical mineral sub-components in GVC's stages from the national customs

Research Assistant | 21st Century China Center 06/2023-09/2023

PI: Prof. Ruixue Jia - Data and Code Exploration Projects funded by 21st Century China Center

- Organized existing data within the professor's research and shared it on the China Data Lab platform
- Explored the application of LLM API for coding textual data, analyzing 30,000 samples of movie synopses using an in-context language model
- Documented code scripts and pulled to GitHub storage, developed plans for the dissemination of research materials

Field Investigator | Survey on Rural Senior Care Services in China, 2021 07/2021-08/2021

PI: Prof. Wenjiong He - Funded by the Ministry of Agriculture and Rural Development of PRC and AIIB, organized by the Research Centre on Aging and Health of Zhejiang University

- Participated in the investigation of 3 counties focusing on elderly services for 65+ in villages totaling 3078 samples
- Helped construct a microdata frame on households and individuals, identified main factors and mechanisms affecting the supply and demand side of elderly services system

Research Assistant | Ecosystem Service Value Evaluation Database Construction 09/2020-01/2021

PI: Prof. Jinxia Wang - Funded by China Center for Agricultural Policy of Peking University

- Reviewed and categorized 170 English articles related to ecosystem service evaluation
- Summarized information indicators of each literature based on their topics, scale, data, and methodology, reported ecosystem services' economic value for each article in the database, assisted to meta-analysis

RESEARCH PROJECT

Capstone Project | School of Global Policy and Strategy & Sempra (SDG&E) 03/2024-06/2024

Advisor: Prof. David Victor & Jim Lambright - Mitigating the Costs of Electrification in California

- Conducted case study and scenario modelling for utilities facing electricity rate hikes under energy transition, integrating insights of optimizing electrification cost distribution across different energy end-users
- Analyzed impacts of demand factors like EV charging, rooftop solar growth, and AI data centers on load patterns and marginal cost, highlighting the potential of battery storage for stabilizing grid during peak demand
- Delivered recommendations backed by demand function estimation and real-world examples ranging from utilizing IRA financial incentives, TOU plan, battery storage construction, and multilateral negotiation

Policy Competition (1st Place) | School of Global Policy and Strategy & Manaus 03/2024-04/2024

Advisor: Prof. Kyle Handley - Strategic Due Diligence for Human Rights in Cotton Supply Chains

- Conducted due diligence analysis for U.S. apparel firms depending on cotton imports from China under the Uyghur Forced Labor Prevention Act (UFLPA)
- Constructed trade-oriented indices to identify countries with high import dependency on Chinese cotton, assessed sourcing risk impacts on firm profit margins, providing guidance for enhancing supply chain resilience
- Delivered recommendations for U.S. apparel firms to mitigate risks, including adjusting import origins, seeking reliable certifications, and diversifying supply chains

Independent Research | School of Global Policy and Strategy 01/2024-04/2024

Advisor: Prof. Ruixue Jia - Impact of Land Tenure Rights on Rural-Urban Migration in China

- Analyzed household-level panel data from the China Health and Retirement Longitudinal Study (CHARLS) using Difference-in-Differences (DID) and two-way fixed effect (TWFE) models
- Found significant effects for land tenure certification on rural-urban migration flow, through providing rural inhabitants with security and economic leverage to seek urban employment
- Provided insights for policy design by exploring the interaction between land security and labor dynamics

Course Project | School of Global Policy and Strategy 01/2024-04/2024

Advisor: Prof. Gordon McCord - Evaluating the Environmental Degradation Effects of Coal Mines in China

- Studied the environmental impacts of coal mining in Inner Mongolia and Shanxi Province, using remote sensing data to analyze changes in vegetation health and water quality
- Utilized Normalized Difference Vegetation Index (NDVI) and Normalized Difference Turbidity Index (NDTI) to quantify vegetation loss and water quality deterioration, revealing significant regional ecological disruptions
- Emphasized tailored environmental regulation frameworks based on sustainable mining practices

Capstone Project | School of Global Policy and Strategy 09/2023-12/2023

Advisor: Prof. Victor Shih - China's Emission Reduction and Energy Efficiency Policies

- Extracted data from China's 13th and 14th Five-Year Plans (FYP), focusing on shifts in emission deduction and energy efficiency strategies, emphasizing the transition from pollution control to carbon neutrality
- Utilized text analysis to evaluate inter-provincial policy heterogeneity, highlighting regional differences in regulatory approaches and their effectiveness
- Researched different environmental regulation patterns across key provinces, providing policy insights to enhance energy efficiency based on regional socioeconomic contexts

Independent Research | Department of Agricultural Economy, Zhejiang University 03/2021-06/2022

Advisor: Prof. Rui Mao - Technological Upgrading on Exports as Shocks to Non-Tariff Barriers

- Utilized trade flow and non-tariff measure (NTM) data from COMTRADE and US FDA, constructing a panel dataset, and using technology complexity index as a proxy of export industry upgrading
- Explored the relationship between technology complexity change and non-tariff measures frequency on the product HS6 granularity through a dynamic GMM model
- Provided trade policy outlook for Chinese domestic entrepreneurs, and induced COVID-19 as an external shock to the global supply chain, developed a DID framework for agricultural products international trade

ADDITIONAL

SKILLS: R; Python; GAMS; STATA; GitHub; SQL; Google Earth Engine (JavaScript); QGIS; ArcGIS; Power BI

LANGUAGES: Mandarin (Native); Cantonese (Professional); English (Professional)